Linear Algebra – MAT 2610

Section 3.2 (Properties of Determinants)

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Determinants and Row Operations

Example: Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & -2 \\ 0 & -2 & 1 \end{bmatrix} \xrightarrow{r_2 \leftrightarrow r_3} \overbrace{\begin{bmatrix} 2 & 1 & 3 \\ 0 & -2 & 1 \\ 0 & 0 & -2 \end{bmatrix}}^{B}$. Calculate $det(A), det(B)$

$$det(A) = a_{11} C_{11} = (2)(-1)^{1+1} det \begin{bmatrix} 0 & -2 \\ -2 & 1 \end{bmatrix} = (2)(1)[0-4] = (2)(-4) = -8$$

$$det(B) = a_{11} C_{11} = (2)(-1)^{1+1} det \begin{bmatrix} -2 & 1 \\ 0 & -2 \end{bmatrix} = (2)(1)(-2)(-2) = 8$$

Determinants and Row Operations

Example: Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & -2 \\ 0 & -2 & 1 \end{bmatrix} \xrightarrow{3r_1 \to r_1} \begin{bmatrix} 6 & 2 & 9 \\ 0 & 0 & -2 \\ 0 & -2 & 1 \end{bmatrix}$. Calculate $det(A)\,, det(B)$

$$det(A) = \underline{\underline{-8}}$$

$$det(B) = a_{11} C_{11} = (6)(-1)^{1+1} det \begin{bmatrix} 0 & -2 \\ -2 & 1 \end{bmatrix}$$

$$= 6(1)[0-4] = \underline{\underline{-24}}$$

Determinants and Row Operations

Example: Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & -2 \\ 0 & -2 & 1 \end{bmatrix} \xrightarrow{r_2 - 2r_1 \to r_2} \begin{bmatrix} 2 & 1 & 3 \\ -4 & -2 & -8 \\ 0 & -2 & 1 \end{bmatrix}$. Calculate $det(A), det(B)$

$det(A) = -8$

$det(B) = a_{11}C_{11} + a_{21}C_{21}$

$= 2(-1)^{1+1} det\begin{bmatrix} -2 & -8 \\ -2 & 1 \end{bmatrix} + (-4)(-1)^{2+1} det\begin{bmatrix} 1 & 3 \\ -2 & 1 \end{bmatrix}$

$= (2)(1)[(-2)(1) - (-8)(-2)] + (-4)(-1)[(1)(1) - (3)(-2)]$

$= (2)(-18) + (4)(7)$

$= -36 + 28 = -8$

Theorem 3

Suppose matrix B is obtained by interchanging two rows of A $(r_i \leftrightarrow r_j)$, then $\det(B) = -\det(A)$

Suppose matrix B is obtained by multiplying one row of A by a scalar $(kr_i \rightarrow r_i)$, then $\det(B) = \mathrm{k} * \det(A)$

Suppose matrix B is obtained by adding a row of A to a multiple of another row $(r_i + kr_j \rightarrow r_i)$, then $\det(B) = \det(A)$

Example: Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 6 & 0 & -2 \\ 0 & -2 & 1 \end{bmatrix}$. Find the echelon form of A and use this to find the determinant of A.

$$A = \begin{bmatrix} 2 & 1 & 3 \\ 6 & 0 & -2 \\ 0 & -2 & 1 \end{bmatrix} \xrightarrow[(1)]{r_2 - 3r_1 \to r_2} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -3 & -11 \\ 0 & -2 & 1 \end{bmatrix} \xrightarrow[(2)]{2r_2 \to r_2} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -6 & -22 \\ 0 & -2 & 1 \end{bmatrix}$$

$$\xrightarrow[(-3)]{-3r_3 \to r_3} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -6 & -22 \\ 0 & 6 & -3 \end{bmatrix} \xrightarrow[(1)]{r_2 + r_3 \to r_3} \begin{bmatrix} 2 & 1 & 3 \\ 0 & -6 & -22 \\ 0 & 0 & -25 \end{bmatrix} = B$$

$$\to \det(A)(1)(2)(-3)(1) = \det(B) \leftarrow$$

$$-6\det(A) = (2)(-6)(-25) = 300$$

$$\det(A) = -50$$

Theorem 4

A square matrix A is invertible if and only if $\det(A) \neq 0$

Example: Let $A = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 0 & -2 \\ 0 & -2 & 1 \end{bmatrix}$. Find $det(A)$, $det(A^T)$

$$det(A) = -8$$

$$A^T = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 0 & -2 \\ 3 & -2 & 1 \end{bmatrix}$$

$$det(A^T) = a_{11} C_{11} = (2)(-1)^{1+1} det \begin{bmatrix} 0 & -2 \\ -2 & 1 \end{bmatrix}$$

$$= (2)(1)[0-4)$$

$$= -8$$

Theorem 5

A square matrix A is invertible, then $\det(A) = \det(A^T)$

Example: Let $A = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} -2 & -1 \\ 3 & 4 \end{bmatrix}$. Find $det(A)$, $det(B)$, $det(AB)$

$det(A) = (2)(1) = 2$

$det(B) = [(-2)(4) - (-1)(3)] = -5$

$$AB = \begin{bmatrix} 2 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -2 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} (2)(-2)+(3)(3) & (2)(-1)+(3)(4) \\ (0)(-2)+(1)(3) & (0)(-1)+(1)(4) \end{bmatrix}$$

$$= \begin{bmatrix} 5 & 10 \\ 3 & 4 \end{bmatrix}$$

$det(AB) = [(5)(4) - (10)(3)] = 20 - 30 = -10$

Note $-10 = (-5)(2)$

$det(A)\, det(B) = det(AB)$

Theorem 6

If A and B are square matrices, then $\det(AB) = \det(A) * \det(B)$

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